

Firestore Push

Overview

Firestore Cloud Messaging integration for push notifications when users are offline or app is in background.

Direction

Outbound — messenger sends push requests to FCM, receives delivery status.

Protocol

- Firestore Cloud Messaging HTTP v1 API
- JSON payload
- Service account authentication

Push Payload

```
{
  "token": "device_fcm_token",
  "notification": {
    "title": "New message",
    "body": "User: Hello!",
    "icon": "/favicon.ico",
    "click_action": "OPEN_CHAT"
  },
  "data": {
    "chatId": "conversation_id",
    "senderId": "user_id",
    "timestamp": "2026-06-20T00:00:00Z"
  }
}
```

Data Flow

```
Message sent → Messenger API → Check if user online
                                ↓ (offline)
                                Send FCM push
                                ↓
                                User receives notification
                                ↓
                                User taps → Open chat
```

Failure Handling

Failure	Response	Recovery
Invalid token	Remove from registry	Clean stale tokens
Rate limited	Retry with backoff	Exponential backoff
Server error	Queue for retry	Error queue + retry job

Related Screens

- [chat](#) — message origin
- [notifications](#) — notification display

Related Integrations

- [auth-service](#) — validates user session for push
- [messenger-api](#) — determines if user is online